

Time Series Practice Questions

1. You are given the following time-series for monthly building settlement (mm):
5, 7, 10, 12, 15, 14, 12, 9, 7, 6, 7, 9
Identify whether each of these components exist in this time series: trend, seasonality and noise. Approximately sketch the decomposition.
2. You are working on modelling displacement of bridge girder. You have collected the following data:

$$t: 0, 1, 2, 3, 4$$

$$x(t): 10, 13, 15, 18, 20$$

Assuming linear trend model $x(t) = a + bt + \epsilon_t$, find the Least-Square Estimate for a and b . What type of error are we implicitly assuming here?

3. Given the following observations:

$$y = Ax + \epsilon$$

$$y = \begin{bmatrix} 5 \\ 8 \\ 6 \end{bmatrix}, A = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \text{cov}(\epsilon) = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{bmatrix}$$

Compute the BLUE of x .

4. In question 3, compute the LS estimate. How do the two compare?
5. You observe the following three time series:
 - 1) $x_t = 5 + \epsilon_t$
 - 2) $x_t = 2t + \epsilon_t$
 - 3) $x_t = 0.7x_{t-1} + \epsilon_t$

Assuming ϵ_t is white noise, determine which of these are stationary.

6. For the process

$$x_t = 3 + \epsilon_t$$

Where ϵ_t is white noise with standard deviation σ^2 , find the mean of x_t , μ and covariance $\text{Cov}(x_t, x_{t+\tau}) = \mathbb{E}((x_t - \mu)(x_{t+\tau} - \mu))$.

7. For the AR(1) model:

$$S_t = 0.6S_{t-1} + \epsilon_t$$

Where ϵ_t is white noise with standard deviation 4. Find the

- a) Is this a stationary time series?
- b) Mean μ
- c) Variance of $S_t = \mathbb{E}((S_t - \mu)(S_t - \mu))$
- d) Lag 1 covariance, $\text{cov}(S_t, S_{t+1}) = \mathbb{E}((S_t - \mu)(S_{t+1} - \mu))$

8. For the given AR(1) model:

$$S_t = 0.8S_{t-1} + \epsilon_t$$

$$S_{10} = 20$$

We know ϵ_t is white noise.

What is the mean of S_{11} , S_{12} and S_{13} ?

9. Using the AR(1) model in Question 8, derive 1-step and 2-step covariances if $\sigma^2 = 9$.